

AlphaForgeBench: Benchmarking End-to-End Trading Strategy Design with Large Language Models

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Abstract

The rapid advancement of Large Language Models (LLMs) has catalyzed the proliferation of diverse financial benchmarks, progressively evolving from static knowledge evaluation to increasingly sophisticated interactive trading simulations. Nevertheless, existing frameworks that assess real-time trading performance largely overlook a fundamental failure mode: the severe behavioral instability exhibited by LLMs in sequential decision-making under financial uncertainty. Through extensive empirical investigation, we demonstrate that when deployed as direct trading agents, LLMs manifest extreme run-to-run variance, produce inconsistent action sequences even under strictly deterministic decoding configurations, and exhibit irrational action flipping across temporally adjacent decision steps. We systematically attribute these pathological behaviors to the models' fundamentally stateless autoregressive architectures, which lack persistent memory of prior actions, and their pronounced sensitivity to continuous-to-discrete action mappings inherent in portfolio allocation tasks. These deficiencies collectively undermine the validity and trustworthiness of numerous existing online and offline financial trading benchmarks, rendering their evaluations unreliable, non-reproducible, and uninformative for meaningful model comparison. To address these limitations, we introduce **ALPHAFORGEBENCH**, a principled evaluation framework that reconceptualizes the role of LLMs from stochastic execution agents to quantitative researchers capable of systematic financial

reasoning. Rather than requiring models to emit discrete trading actions, **ALPHAFORGEBENCH** tasks LLMs with generating executable alpha factors and composing factor-based trading strategies grounded in financial domain knowledge. This paradigm shift decouples reasoning from execution mechanics, enabling fully deterministic and reproducible evaluation while maintaining close alignment with real-world quantitative research workflows. Extensive experiments across multiple state-of-the-art LLMs demonstrate that **ALPHAFORGEBENCH** effectively eliminates execution-induced instability, yields highly reproducible outcomes, and provides a rigorous and discriminative benchmark for assessing LLMs' capacity for financial reasoning, strategy formulation, and alpha discovery. Webpage at <https://finbrain-lab-hkustgz.github.io/AlphaForgeBench>.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Large Language Models, Alpha Factor Discovery, Quantitative Finance, Factor-Based Trading Strategies, Benchmarking

ACM Reference Format:

Wentao Zhang, Mingxuan Zhao, Jincheng Gao, Jieshun You, Huaiyu Jia, Yilei Zhao, Bo An, and Shuo Sun. 2026. AlphaForgeBench: Benchmarking End-to-End Trading Strategy Design with Large Language Models. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26)*, August 09–13, 2026, Jeju Island, Republic of Korea. ACM, New York, NY, USA, 80 pages. <https://doi.org/10.1145/3770855.3817500>

1 Introduction

The rapid advancement of large language models (LLMs) has spurred the development of numerous benchmarks to assess model capabilities across diverse domains. In the financial domain, early benchmarks focused on evaluating general financial knowledge through

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ACM ISBN 979-8-4007-2259-2/2026/08

<https://doi.org/10.1145/3770855.3817500>

question-answering (QA) tasks, including numerical reasoning over financial reports (e.g., FinQA [6], TAT-QA [28]), conversational finance QA (e.g., ConvFinQA [7]), and comprehensive multi-task evaluation frameworks (e.g., BloombergGPT [18], FinGPT [11], PIXIU [21], FinBen [19]). However, these benchmarks primarily measure *encyclopedic knowledge* and *static reasoning* over historical snapshots, which fail to capture an LLM's ability to make sequential trading decisions under non-stationary market conditions. As the field evolved, researchers shifted toward evaluating LLMs' real-time trading capabilities, with benchmarks such as Alpha Arena [2] establishing live trading evaluation paradigms that assess LLM agents' adaptability and alpha-seeking capabilities directly within dynamic, executing financial markets. While these trading benchmarks have expanded the scope of financial LLMs evaluation, they have largely overlooked a critical issue: LLMs exhibit extreme *instability* in their performance on financial trading tasks.

Specifically, when LLMs or LLM-based agents directly emit trading decisions (e.g., BUY/HOLD/SELL signals), their outputs demonstrate severe instability across multiple dimensions. **(1) Run-to-run variance in performance metrics.** Under identical settings, the same LLM produces dramatically different trading trajectories across multiple runs on the same financial data (e.g., OHLCV time series with technical indicators), resulting in substantial variance in returns, drawdowns, and other performance metrics. **(2) Inconsistent action sequences even under deterministic decoding.** Even with $\text{temperature}=0$ (deterministic decoding), LLMs generate completely different trading action sequences across runs on identical market data, exhibiting no consistency in decision-making patterns. **(3) Rapid action flipping persists despite hard constraints.** LLMs exhibit a tendency to rapidly flip trading actions (e.g., buying immediately after selling, or selling immediately after buying) when performing trading tasks. Even when explicit behavioral guardrails are incorporated into prompts (e.g., minimum holding periods, cooldown windows, or historical action sequences), these prompt-based constraints cannot fully eliminate such rapid flipping behavior. Detailed analyses are provided in Appx. C.

Fundamentally, this phenomenon can be primarily attributed to the following three aspects. First, LLMs are inherently stateless, instantaneous decision-making models. In trading tasks, each action output is an independent re-evaluation based on a "current input snapshot." The model possesses no natural memory of its recent buy or sell executions, nor does it view position holding as a long-term state requiring consistency. Consequently, even slight variations in market features in the subsequent step can lead the model to generate divergent or even diametrically opposite actions. Second, LLMs are highly sensitive to the mapping of continuous market signals to discrete actions. While inputs such as prices and technical indicators vary continuously, the outputs are discrete actions. This continuous-to-discrete mapping amplifies the impact of minor fluctuations, causing the model to abruptly shift its stance upon slight indicator reversals or semantic shifts. Crucially, it lacks the mechanisms of inertia, tolerance intervals, and strategic waiting typically inherent in professional trading. Third, LLMs fundamentally perform classification rather than strategy optimization. Directly prompting an LLM to output an action essentially tasks it with assigning a "most reasonable action" label to the current market state, rather than maximizing long-term returns

within a sequential decision-making framework subject to costs, positions, and time constraints. Because the model is agnostic to transaction fees, slippage, and penalties for excessive or insufficient trading, it is highly prone to collapsing into extreme patterns of either over-trading or complete inactivity.

To address these limitations, we propose a new evaluation paradigm that shifts from black-box trading to white-box logic generation. We introduce **ALPHAFORGE BENCH**, a comprehensive benchmark designed to assess LLMs on their ability to generate executable alpha factors and trading strategy code, effectively positioning the LLM as a quantitative researcher rather than a stochastic execution agent. This paradigm shift offers three critical advantages that directly mitigate the aforementioned instability. First, it fundamentally solves the continuous-to-discrete sensitivity problem by decoupling reasoning from execution. By compelling the LLM to formalize its decision boundaries into explicit algorithmic rules, the stochastic nature of the model is confined to the generation phase, rendering the subsequent execution strictly deterministic and immune to the random action flipping observed in direct-trading tasks. Second, it resolves the state management issue. Unlike stateless models that struggle to maintain continuity, the generated code inherently preserves internal states and logic across the entire time series to ensure consistent decision-making. Third, this setup mirrors the real-world workflow of quantitative finance where researchers synthesize strategies and engines execute them. This alignment enables transparent logic verification and provides a rigorous metric to assess whether the model has truly learned financial reasoning or is merely overfitting to market noise. Our contributions are summarized as follows:

- We systematically analyze the critical flaws in existing financial trading benchmarks. We demonstrate that the inherent instability of LLMs in direct-trading tasks renders traditional performance metrics unreliable, preventing an accurate assessment of true financial reasoning capabilities.
- We introduce **ALPHAFORGE BENCH**, a novel benchmark that repositions LLMs from stochastic agents to quantitative researchers. By evaluating the generation of executable alpha factors and strategy code, our framework provides a deterministic and robust metric for measuring financial logic synthesis.
- We conduct extensive experiments across state-of-the-art LLMs to validate the effectiveness of our approach. The results confirm that **ALPHAFORGE BENCH** offers a significantly more stable and discriminatory assessment of financial capabilities compared to direct-trading baselines.

2 Related Work

2.1 Financial Knowledge Benchmark

Recent advancements in financial question answering have driven the development of diverse benchmarks ranging from specific reasoning tasks to holistic system evaluations. Early efforts focused on numerical and hybrid reasoning, with TAT-QA [28] addressing multi-step reasoning over tabular-textual data, FinQA [6] targeting numerical reasoning on financial reports, and ConvFinQA [7] extending this to conversational contexts. As LLMs evolved, researchers introduced comprehensive frameworks to evaluate broader capabilities: BloombergGPT [18] validated the efficacy of

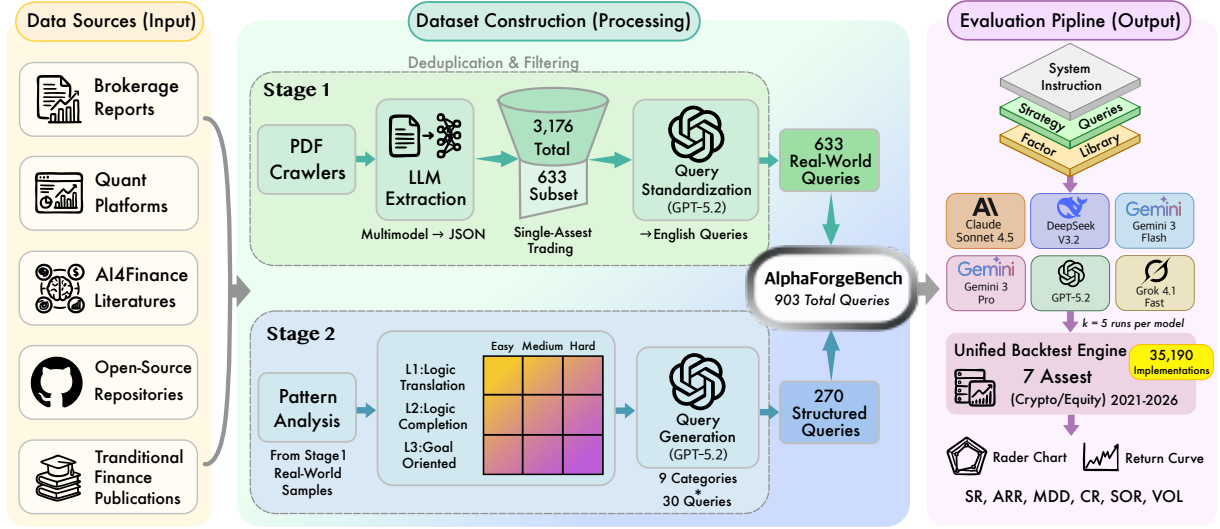


Figure 1: The framework of ALPHAForgeBench.

domain-specific pretraining, FinGPT [11] democratized internet-scale financial data for open-source training, PIXIU [21] established a multi-task instruction tuning benchmark, and FinBen [19] offered a holistic evaluation across seven dimensions of financial intelligence. Addressing the complexity of long-form generation, FinTextQA [4] utilized RAG-based metrics for extensive textbook-level queries. Furthermore, recent works have tailored benchmarks to specific linguistic and user needs, with CFinBench [14] and FinEva [3] constructing fine-grained evaluation systems for Chinese financial knowledge, while UCFE [23] pioneered a user-centric framework to align model performance with dynamic human preferences across diverse roles. However, these existing frameworks remain largely confined to passive knowledge retrieval and phenomenon explication, failing to bridge the critical gap between theoretical financial understanding and actionable, strategy-driven trading execution in practice.

2.2 Financial Trading Benchmark

The evolution of financial LLMs has transitioned from passive analysis to autonomous decision-making agents, necessitating rigorous benchmarks for trading and forecasting. In terms of agentic architecture, FINCON [25] introduces a hierarchical multi-agent system with conceptual verbal reinforcement for risk-aware strategies, while AlphaFin [5] establishes a retrieval-augmented pipeline to evaluate end-to-end fundamental reasoning and alpha generation. [12, 20] analyze the performance of the large model in terms of price prediction and returns. To standardize decision-making assessments, INVESTORBENCH [10] provides a multi-asset simulation environment, revealing that agents struggle to outperform buy-and-hold baselines. Critically, addressing the “time-travel” bias in historical backtesting, recent works have pivoted toward live-market evaluation: DeepFund [9] demonstrates that even SOTA models incur losses in real-time fund management. FutureX [15, 26] pioneers a live, anti-contamination forecasting framework to evaluate the real-time predictive intelligence of LLM agents on unfolding global events. To eliminate look-ahead bias and bridge the gap between simulation and reality, Alpha Arena [2], RockAlpha [16],

and LiveTradeBench [24] collectively establish a live trading benchmark paradigm, rigorously evaluating the real-time adaptability and alpha-seeking capabilities of LLM agents directly within dynamic, executing financial markets.

However, these live evaluation paradigms introduce a fundamental *reproducibility crisis*: evaluation results are bound to the specific execution time window and cannot be independently replicated, a problem further compounded by the inherent stochasticity of LLMs, which produce drastically different trading actions across runs even under deterministic decoding (Section C), rendering single-run live evaluations statistically unreliable. While existing benchmarks have advanced the evaluation of autonomous trading agents, they overlook the necessity of a holistic alpha mining pipeline that integrates strategy formulation, executable code generation, and rigorous backtesting, and largely neglect quantifying the inherent stochasticity and instability of LLM-driven financial decision-making.

3 AlphaForgeBench

In this section we present **ALPHAForgeBench**, a benchmark that evaluates LLMs as quantitative researchers who synthesize executable trading strategies rather than emit point-wise trading actions. The overall framework is illustrated in Figure 1 and is organized along three axes. First, we describe the *dataset construction* process (Section 3.1), which proceeds in two stages: Stage 1 collects natural-language queries together with their ground-truth alpha factors and trading strategies from diverse real-world sources; Stage 2 draws on the patterns and complexity profiles observed in these real-world samples to systematically generate augmented queries across a 3×3 level-grade difficulty taxonomy via LLMs, combining authenticity with controlled diagnostic granularity. Second, we detail the *evaluation pipeline* (Section 3.2), in which each query is fed to the evaluated LLM to produce executable factor and strategy code, and the generated code is then executed within a standardized backtest engine across multiple assets and market regimes to yield quantitative performance profiles. Third, we introduce the

evaluation methodology (Section 3.3), comprising financial performance metrics and statistical protocols that assess both the absolute quality and the cross-run stability of LLM-generated strategies.

3.1 Dataset Construction

The construction of **ALPHAFORGE BENCH** follows a two-stage pipeline that combines the ecological validity of real-world strategies with the diagnostic precision of synthetically structured queries.

Stage 1: Real-world Strategy Collection. We curate a diverse corpus of alpha factors and factor-based trading strategies from five complementary source categories: brokerage research reports, quantitative investment platforms (WorldQuant [17], JoinQuant [8]), AI-in-finance literature, open-source repositories (Qlib [22], OpenFE [27]), and traditional finance publications. We build an automated extraction agent, powered by *gemini-3-flash-preview*, that ingests each collected document and produces structured records comprising factor names, mathematical definitions, trading logic, and financial rationale. After deduplication and quality filtering, Stage 1 yields **3,176** factor-strategy entries spanning three strategy types: single-asset trading (633), portfolio management (2,172), and multi-asset trading (371). In this work, we restrict evaluation to the **633** single-asset subset to isolate the LLM’s signal-generation capability from confounding portfolio-construction effects, thereby serving as an ecologically valid baseline for gauging benchmark difficulty. This restriction is a deliberate design choice rather than a limitation of scope: the single-asset setting provides a cleaner, more controlled evaluation environment that targets core signal-generation and rule-construction capabilities without the additional complexity introduced by asset allocation, cross-asset dependencies, and portfolio-level risk constraints. The portfolio management (2,172) and multi-asset trading (371) subsets are reserved for systematic evaluation in future work, along with the corresponding backtesting infrastructure already developed. Full extraction prompts and dataset statistics are provided in Section D.

Stage 2: LLM-augmented Structured Query Generation. While Stage 1 offers ecologically valid test cases, its query distribution is non-uniform across difficulty and is not tailored for controlled diagnosis of specific cognitive demands. Stage 2 therefore constructs **270** additional benchmark queries under a 3×3 *level-grade* taxonomy, grounded in the strategy patterns and complexity profiles observed in the real-world collection. The three *levels* isolate distinct strategy-generation skills: **Level 1** (Logic Translation) provides fully specified if-then rules to test faithful code translation; **Level 2** (Logic Completion) supplies strategic skeletons with critical parameters omitted, requiring domain-grounded inference; and **Level 3** (Goal-Oriented Generation) specifies only high-level investment objectives, demanding end-to-end strategy design from first principles. Orthogonally, three *grades* (Easy, Medium, Hard) modulate complexity via the number of conditions, the degree of underspecification, and the depth of state-dependent control flow, yielding nine fine-grained difficulty cells (Appx. D).

In summary, the **ALPHAFORGE BENCH** query set comprises two complementary components: **633** real-world queries from Stage 1 that ensure ecological validity, and **270** difficulty-specialized queries from Stage 2 that enable controlled, fine-grained diagnostic evaluation across the full 3×3 level-grade grid, providing balanced

coverage and facilitating stratified analyses of model failure modes under varying cognitive demands.

3.2 Evaluation Pipeline

Given the curated query set described above (633 real-world queries from Stage 1 and 270 difficulty-specialized queries from Stage 2), **ALPHAFORGE BENCH** evaluates each model via a standardized generate-and-backtest pipeline (Figure 1), comprising prompt instantiation, code synthesis, and backtest-based assessment.

Step 1: Prompt construction. Each benchmark query is assembled into a standardized prompt comprising three semantically distinct components (see the *Evaluation Pipeline* panel of Figure 1): (i) a *system instruction* that defines the code-generation task, specifies the available data schema (open-high-low-close-volume (OHLCV) columns augmented with precomputed technical-indicator factors), and prescribes the expected output interface; (ii) the *strategy query*, i.e., the natural-language description of the target trading strategy drawn from either Stage 1 or Stage 2; and (iii) a *factor-library reference* that enumerates all supported indicator names together with their formal mathematical definitions, providing the model with a complete and unambiguous specification of the default feature space. Notably, the model is not restricted to this predefined indicator set: if a strategy requires novel factors, the model may generate the corresponding factor-computation code, which our backtest engine dynamically registers and incorporates into the evaluation, thereby granting models the flexibility to extend the feature space on the fly. The prompt template is held strictly identical across all evaluated models, ensuring that any observed performance differences can be attributed solely to model capabilities rather than prompt-engineering artifacts.

Step 2: Code generation. The assembled prompt is dispatched to each evaluated LLM through its official API. The model must return a self-contained Python function (`generate_signal`) that consumes a dataframe of OHLCV columns and precomputed indicator factors and produces a trading-signal series dictating position actions (e.g., invest versus hold cash). We enforce strict conformance to the backtest engine’s interface contract (function signature, permitted column references, and output format), enabling fully automated execution without manual intervention.

Step 3: Backtest-based assessment. Each generated implementation is executed within a unified, deterministic backtest engine on historical daily price data spanning seven assets across two market regimes: cryptocurrency and US equity. The engine computes a suite of standard financial metrics covering return generation, risk exposure, and risk-adjusted efficiency, producing fully reproducible quantitative profiles that support systematic comparison across models, query sources, and the nine Stage 2 difficulty cells. This deterministic design ensures that any observed cross-run variance originates exclusively from the inherent stochasticity of LLM generation rather than evaluation-side randomness.

3.3 Evaluation Methodology

We evaluate LLM-generated strategies along three complementary dimensions. First, each backtest run yields a suite of standard financial metrics spanning return generation, risk exposure, and risk-adjusted efficiency; to account for the stochasticity of LLM

generation, all metrics are reported as mean $\pm$ standard deviation over multiple independent runs, capturing both expected performance and generation stability. Second, results are organized into stratified tables along three axes: (i) overall model ranking aggregated across all queries and assets, (ii) per-asset decomposition over the 7 backtest assets to quantify cross-market generalization from cryptocurrency to US equity, and (iii) per-level decomposition (Stage 2 only) by the three difficulty levels of the 3×3 taxonomy to reveal how capabilities degrade under increasing cognitive demands. Third, tabular results are complemented by radar charts that render each model’s multi-metric profile for intuitive risk-return comparison, grouped bar charts and box plots that expose cross-asset robustness and inter-model dispersion, and cumulative return curves that surface temporal dynamics such as divergence during market stress and convergence in calm regimes.

4 Experiments

We validate **ALPHAForgeBENCH** along two complementary evaluation tracks that mirror the two-stage dataset construction. **Track 1** (real-world queries) evaluates all 633 single-asset queries from Stage 1 to establish ecological validity, baseline difficulty calibration, and a consistency check against the structured track. **Track 2** (structured queries) evaluates the 270 queries from Stage 2 organized by the 3×3 level-grade taxonomy, enabling fine-grained diagnosis of model strengths, weaknesses, and specific failure modes across difficulty dimensions. This dual-track design also serves as an empirical bias-mitigation mechanism: concordance between model rankings on Stage 1 (real-world) and Stage 2 (LLM-augmented) queries validates that the structured generation process preserves genuine capability differences rather than introducing systematic distributional bias favoring particular models.

4.1 Experimental Settings

Evaluated models. We benchmark six frontier LLMs spanning five providers: *claude-sonnet-4.5*, *deepseek-v3.2*, *gemini-3-flash-preview*, *gemini-3-pro-preview*, *gpt-5.2*, and *grok-4.1-fast*. All models are queried through their official APIs under identical prompt templates with no model-specific tuning.

Generation protocol. Every experiment runs $k=5$ independent generations per query to quantify run-to-run variability. Stage 1 uses $\tau=0.7$ (633 queries $\times$ 6 models $\times$ 5 runs = 18,990 samples); Stage 2 is evaluated at both $\tau=0.7$ and $\tau=0$ (greedy) for a temperature ablation (270 queries $\times$ 6 models $\times$ 2 temperatures $\times$ 5 runs = 16,200 samples). The grand total is **35,190** generated strategy implementations. All metrics are reported as mean $\pm$ std.

Backtest configuration. Each generated strategy is executed within the unified backtest engine across **7 assets** spanning two distinct market regimes: 2 cryptocurrencies (BTCUSDT, ETHUSDT; sourced from Binance) and 5 US equities (AAPL, GOOGL, MSFT, NVDA, TSLA; sourced from Yahoo Finance). The evaluation window spans **5 years** (2021-01-01 to 2026-01-01), deliberately selected to cover heterogeneous market conditions including bull rallies, bear corrections, AI-driven recoveries, and prolonged consolidation phases, ensuring that no single strategy style is systematically favored. We adopt a controlled execution protocol with daily data frequency, a 300-day lookback window, long-only single-asset semantics (binary invest/cash signal), and a fixed one-way transaction

cost of 10^{-3} , thereby isolating intrinsic signal quality from confounding portfolio-construction effects. The long-only constraint is adopted for cross-market consistency, as short-selling is subject to asymmetric restrictions and costs across equity and cryptocurrency markets. Slippage and liquidity constraints are intentionally excluded: these factors are highly dependent on market microstructure (e.g., order-book depth and execution mechanisms) and vary substantially across assets and trading frequencies; introducing a unified slippage model would add evaluation-side uncertainty without improving the fairness of cross-model comparisons. Full parameter justifications are provided in Appx. E.

Evaluation metrics. We assess each strategy using six standard financial metrics: Annual Rate of Return (**ARR**), Sharpe Ratio (**SR**), Maximum Drawdown (**MDD**), Calmar Ratio (**CR**), Sortino Ratio (**SoR**), and Volatility (**VOL**). These jointly capture return generation (ARR), risk exposure (MDD, VOL), and risk-adjusted efficiency (SR, CR, SoR), providing a multi-dimensional profile of strategy quality. Formal definitions are given in Appx. E.

4.2 Results on Real-world Queries

Table 1: Results on real-world queries (mean $\pm$ std, 633 queries), stratified by overall (green) and asset (gray). Best per block in bold; $\uparrow/\downarrow$: higher/lower is better.

Model	SR $\uparrow$	ARR $\uparrow$	MDD $\downarrow$	CR $\uparrow$	SoR $\uparrow$	VOL $\downarrow$
Overall						
<i>claude-sonnet-4.5</i>	0.378 $\pm$ 0.268	0.138 $\pm$ 0.122	0.138 $\pm$ 0.122	1.456 $\pm$ 1.106	0.636 $\pm$ 0.488	0.187 $\pm$ 0.165
<i>deepseek-v3.2</i>	0.329 $\pm$ 0.272	0.116 $\pm$ 0.122	0.114 $\pm$ 0.120	1.575 $\pm$ 1.227	0.548 $\pm$ 0.494	0.155 $\pm$ 0.163
<i>gemini-3-flash-preview</i>	0.388 $\pm$ 0.268	0.142 $\pm$ 0.122	0.138 $\pm$ 0.119	1.504 $\pm$ 1.131	0.648 $\pm$ 0.488	0.189 $\pm$ 0.161
<i>gemini-3-pro-preview</i>	0.449 $\pm$ 0.262	0.171 $\pm$ 0.123	0.174 $\pm$ 0.119	1.411 $\pm$ 1.165	0.767 $\pm$ 0.493	0.237 $\pm$ 0.162
<i>gpt-5.2</i>	0.342 $\pm$ 0.279	0.123 $\pm$ 0.119	0.122 $\pm$ 0.118	1.534 $\pm$ 1.386	0.575 $\pm$ 0.503	0.166 $\pm$ 0.161
<i>grok-4.1-fast</i>	0.366 $\pm$ 0.276	0.135 $\pm$ 0.122	0.142 $\pm$ 0.124	1.396 $\pm$ 1.038	0.618 $\pm$ 0.500	0.192 $\pm$ 0.168
BTCUSDT (Cryptocurrency)						
<i>claude-sonnet-4.5</i>	0.279 $\pm$ 0.234	0.110 $\pm$ 0.099	0.189 $\pm$ 0.154	0.732 $\pm$ 0.792	0.490 $\pm$ 0.398	0.213 $\pm$ 0.175
<i>deepseek-v3.2</i>	0.239 $\pm$ 0.231	0.094 $\pm$ 0.097	0.155 $\pm$ 0.155	0.861 $\pm$ 0.967	0.421 $\pm$ 0.393	0.175 $\pm$ 0.175
<i>gemini-3-flash-preview</i>	0.294 $\pm$ 0.225	0.115 $\pm$ 0.096	0.193 $\pm$ 0.152	0.800 $\pm$ 0.865	0.511 $\pm$ 0.380	0.218 $\pm$ 0.171
<i>gemini-3-pro-preview</i>	0.338 $\pm$ 0.223	0.131 $\pm$ 0.097	0.241 $\pm$ 0.149	0.628 $\pm$ 0.552	0.592 $\pm$ 0.376	0.272 $\pm$ 0.167
<i>gpt-5.2</i>	0.255 $\pm$ 0.230	0.098 $\pm$ 0.097	0.170 $\pm$ 0.152	0.719 $\pm$ 0.713	0.446 $\pm$ 0.394	0.192 $\pm$ 0.172
<i>grok-4.1-fast</i>	0.273 $\pm$ 0.231	0.104 $\pm$ 0.096	0.196 $\pm$ 0.160	0.701 $\pm$ 0.782	0.481 $\pm$ 0.389	0.220 $\pm$ 0.179
ETHUSDT (Cryptocurrency)						
<i>claude-sonnet-4.5</i>	0.260 $\pm$ 0.232	0.119 $\pm$ 0.149	0.205 $\pm$ 0.208	0.778 $\pm$ 0.626	0.398 $\pm$ 0.359	0.242 $\pm$ 0.244
<i>deepseek-v3.2</i>	0.220 $\pm$ 0.233	0.103 $\pm$ 0.147	0.169 $\pm$ 0.205	0.851 $\pm$ 0.669	0.336 $\pm$ 0.361	0.200 $\pm$ 0.241
<i>gemini-3-flash-preview</i>	0.260 $\pm$ 0.236	0.121 $\pm$ 0.151	0.206 $\pm$ 0.203	0.737 $\pm$ 0.606	0.400 $\pm$ 0.363	0.242 $\pm$ 0.239
<i>gemini-3-pro-preview</i>	0.306 $\pm$ 0.224	0.137 $\pm$ 0.152	0.263 $\pm$ 0.198	0.633 $\pm$ 0.558	0.475 $\pm$ 0.344	0.307 $\pm$ 0.232
<i>gpt-5.2</i>	0.219 $\pm$ 0.218	0.084 $\pm$ 0.106	0.180 $\pm$ 0.198	0.675 $\pm$ 0.641	0.336 $\pm$ 0.333	0.211 $\pm$ 0.232
<i>grok-4.1-fast</i>	0.255 $\pm$ 0.233	0.112 $\pm$ 0.145	0.218 $\pm$ 0.209	0.645 $\pm$ 0.564	0.392 $\pm$ 0.358	0.254 $\pm$ 0.244
AAPL (US Equity)						
<i>claude-sonnet-4.5</i>	0.757 $\pm$ 0.533	0.151 $\pm$ 0.118	0.079 $\pm$ 0.059	2.282 $\pm$ 1.784	1.068 $\pm$ 0.771	0.123 $\pm$ 0.089
<i>deepseek-v3.2</i>	0.704 $\pm$ 0.546	0.138 $\pm$ 0.118	0.069 $\pm$ 0.060	2.529 $\pm$ 2.036	0.994 $\pm$ 0.782	0.108 $\pm$ 0.089
<i>gemini-3-flash-preview</i>	0.756 $\pm$ 0.522	0.151 $\pm$ 0.116	0.081 $\pm$ 0.058	2.252 $\pm$ 1.910	1.067 $\pm$ 0.756	0.126 $\pm$ 0.087
<i>gemini-3-pro-preview</i>	0.805 $\pm$ 0.500	0.162 $\pm$ 0.113	0.094 $\pm$ 0.059	1.954 $\pm$ 1.431	1.138 $\pm$ 0.738	0.143 $\pm$ 0.087
<i>gpt-5.2</i>	0.668 $\pm$ 0.547	0.132 $\pm$ 0.120	0.071 $\pm$ 0.061	2.273 $\pm$ 1.880	0.944 $\pm$ 0.794	0.110 $\pm$ 0.092
<i>grok-4.1-fast</i>	0.686 $\pm$ 0.545	0.138 $\pm$ 0.119	0.077 $\pm$ 0.062	2.095 $\pm$ 1.709	0.973 $\pm$ 0.786	0.119 $\pm$ 0.092
GOOGL (US Equity)						
<i>claude-sonnet-4.5</i>	0.657 $\pm$ 0.539	0.190 $\pm$ 0.203	0.121 $\pm$ 0.121	2.374 $\pm$ 2.209	1.177 $\pm$ 1.024	0.184 $\pm$ 0.174
<i>deepseek-v3.2</i>	0.541 $\pm$ 0.532	0.151 $\pm$ 0.191	0.099 $\pm$ 0.116	2.204 $\pm$ 1.988	0.959 $\pm$ 1.008	0.150 $\pm$ 0.169
<i>gemini-3-flash-preview</i>	0.656 $\pm$ 0.539	0.188 $\pm$ 0.205	0.124 $\pm$ 0.123	2.335 $\pm$ 2.279	1.170 $\pm$ 1.023	0.190 $\pm$ 0.176
<i>gemini-3-pro-preview</i>	0.752 $\pm$ 0.525	0.225 $\pm$ 0.209	0.105 $\pm$ 0.129	2.341 $\pm$ 2.210	1.390 $\pm$ 1.011	0.229 $\pm$ 0.183
<i>gpt-5.2</i>	0.602 $\pm$ 0.556	0.179 $\pm$ 0.206	0.109 $\pm$ 0.121	2.502 $\pm$ 2.422	1.089 $\pm$ 1.056	0.167 $\pm$ 0.175
<i>grok-4.1-fast</i>	0.617 $\pm$ 0.542	0.175 $\pm$ 0.203	0.121 $\pm$ 0.128	2.331 $\pm$ 2.231	1.113 $\pm$ 1.028	0.183 $\pm$ 0.184
MSFT (US Equity)						
<i>claude-sonnet-4.5</i>	0.142 $\pm$ 0.248	0.020 $\pm$ 0.036	0.091 $\pm$ 0.068	0.319 $\pm$ 0.703	0.231 $\pm$ 0.285	0.121 $\pm$ 0.088
<i>deepseek-v3.2</i>	0.133 $\pm$ 0.242	0.020 $\pm$ 0.034	0.079 $\pm$ 0.067	0.352 $\pm$ 0.776	0.214 $\pm$ 0.280	0.105 $\pm$ 0.088
<i>gemini-3-flash-preview</i>	0.147 $\pm$ 0.250	0.021 $\pm$ 0.036	0.091 $\pm$ 0.066	0.289 $\pm$ 0.662	0.235 $\pm$ 0.288	0.121 $\pm$ 0.086
<i>gemini-3-pro-preview</i>	0.176 $\pm$ 0.253	0.024 $\pm$ 0.037	0.102 $\pm$ 0.067	0.316 $\pm$ 0.620	0.268 $\pm$ 0.290	0.137 $\pm$ 0.086
<i>gpt-5.2</i>	0.125 $\pm$ 0.245	0.018 $\pm$ 0.034	0.081 $\pm$ 0.070	0.293 $\pm$ 0.608	0.204 $\pm$ 0.284	0.108 $\pm$ 0.091
<i>grok-4.1-fast</i>	0.138 $\pm$ 0.232	0.019 $\pm$ 0.034	0.086 $\pm$ 0.069	0.306 $\pm$ 0.665	0.217 $\pm$ 0.273	0.115 $\pm$ 0.090
NVDA (US Equity)						
<i>claude-sonnet-4.5</i>	0.289 $\pm$ 0.324	0.090 $\pm$ 0.135	0.133 $\pm$ 0.153	1.071 $\pm$ 1.506	0.558 $\pm$ 0.635	0.207 $\pm$ 0.233
<i>deepseek-v3.2</i>	0.242 $\pm$ 0.315	0.074 $\pm$ 0.128	0.110 $\pm$ 0.149	1.090 $\pm$ 1.522	0.469 $\pm$ 0.618	0.171 $\pm$ 0.227
<i>gemini-3-flash-preview</i>	0.300 $\pm$ 0.425	0.304 $\pm$ 0.383	0.138 $\pm$ 0.168	2.781 $\pm$ 2.794	0.572 $\pm$ 0.713	0.215 $\pm$ 0.256
<i>gemini-3-pro-preview</i>	0.372 $\pm$ 0.318	0.112 $\pm$ 0.138	0.184 $\pm$ 0.162	0.939 $\pm$ 1.314	0.739 $\pm$ 0.650	0.284 $\pm$ 0.244
<i>gpt-5.2</i>	0.272 $\pm$ 0.315	0.086 $\pm$ 0.132	0.122 $\pm$ 0.151	1.094 $\pm$ 1.355	0.532 $\pm$ 0.634	0.190 $\pm$ 0.229
<i>grok-4.1-fast</i>	0.293 $\pm$ 0.327	0.088 $\pm$ 0.130	0.139 $\pm$ 0.155	0.952 $\pm$ 1.262	0.563 $\pm$ 0.634	0.214 $\pm$ 0.235
TSLA (US Equity)						
<i>claude-sonnet-4.5</i>	0.274 $\pm$ 0.420	0.289 $\pm$ 0.381	0.140 $\pm$ 0.174	2.611 $\pm$ 2.772	0.545 $\pm$ 0.713	0.216 $\pm$ 0.264
<i>deepseek-v3.2</i>	0.240 $\pm$ 0.393	0.239 $\pm$ 0.359	0.115 $\pm$ 0.164	2.629 $\pm$ 2.745	0.466 $\pm$ 0.680	0.179 $\pm$ 0.251
<i>gemini-3-flash-preview</i>	0.300 $\pm$ 0.425	0.304 $\pm$ 0.383	0.138 $\pm$ 0.168	2.781 $\pm$ 2.794	0.572 $\pm$ 0.713	0.215 $\pm$ 0.256
<i>gemini-3-pro-preview</i>	0.396 $\pm$ 0.444	0.409 $\pm$ 0.400	0.186 $\pm$ 0.175	2.756 $\pm$ 2.738	0.770 $\pm$ 0.738	0.290 $\pm$ 0.264
<i>gpt-5.2</i>	0.275 $\pm$ 0.431	0.272 $\pm$ 0.379	0.120 $\pm$ 0.163	2.972 $\pm$ 3.080	0.517 $\pm$ 0.710	0.189 $\pm$ 0.251
<i>grok-4.1-fast</i>	0.303 $\pm$ 0.422	0.315 $\pm$ 0.383	0.153 $\pm$ 0.173	2.659 $\pm$ 2.676	0.597 $\pm$ 0.713	0.237 $\pm$ 0.262

4.2.1 Overall model comparison. As shown in the Overall block of Table 1, three salient observations emerge. **(1) A clear performance hierarchy exists on return-oriented metrics.** Models separate into three tiers: *gemini-3-pro-preview* leads decisively, followed by a middle tier (*gemini-3-flash-preview*, *claude-sonnet-4.5*), with the remaining models trailing. The 5.5% ARR spread between the best and worst models translates to a 47% relative improvement, which is economically meaningful over the 5-year evaluation horizon. **(2) Return and risk rankings are inversely correlated,** suggesting that LLMs encode distinct implicit risk preferences in their generated code. Notably, *gemini-3-pro-preview* consistently favors aggressive, high-conviction signal logic (highest SR and ARR, but also highest MDD and VOL), whereas *deepseek-v3.2* converges on conservative, risk-controlled strategies (lowest MDD and VOL, yet highest CR). This return-risk inversion underscores the necessity of multi-metric evaluation: a single metric cannot capture the full spectrum of model behavior. **(3) Observed performance gaps are statistically reliable.** The intra-query standard deviation across 5 runs is typically an order of magnitude smaller than the inter-query standard deviation, confirming that ranking differences reflect genuine capability gaps rather than generation noise. Detailed distributional and visual analyses are provided in Appx. F.

4.2.2 Per-asset analysis. Disaggregating by asset in Table 1 yields three further observations. **(1) A consistent difficulty gradient emerges across assets.** Trend-rich US large-caps (AAPL, GOOGL) are the easiest targets for all models, cryptocurrencies (BTCUSDT, ETHUSDT) occupy an intermediate tier with higher volatility and deeper drawdowns, and MSFT proves the hardest asset, likely due to its narrower trading ranges during the evaluation period. **(2) High-volatility assets amplify inter-model dispersion.** TSLA exhibits both the highest absolute returns and the widest cross-model variance, indicating that extreme and highly non-stationary price dynamics effectively magnify differences in the signal logic generated by each LLM. **(3) Model rankings are robust across asset classes.** The relative ordering identified in the aggregate analysis is preserved across all seven assets: return-oriented leaders and risk-oriented leaders maintain their respective positions regardless of whether the market is cryptocurrency or US equity. This cross-asset stability strongly suggests that the implicit risk preferences encoded by each LLM are intrinsic to its own generation behavior rather than artifacts of any particular market environment.

4.2.3 Findings and conclusions. Synthesizing the above analyses, the Stage 1 evaluation yields four principal findings. **(1) High code-generation reliability:** all six frontier LLMs produce executable strategy code with >96% backtest pass rates, establishing a solid foundation for the code-generation evaluation paradigm. **(2) Stable and reproducible evaluation:** the intra-query (run-to-run) variance is an order of magnitude smaller than the inter-query variance, confirming that the code-generation paradigm confines LLM stochasticity to a single generation step while guaranteeing deterministic execution thereafter, in stark contrast to direct-trading benchmarks where run-to-run variance routinely exceeds inter-model variance. **(3) Implicit risk preferences:** as revealed by the return-risk inversion in the overall comparison, different LLMs encode distinct and stable risk personalities in their generated code, and these characteristic risk profiles persist across all runs, assets, and

metrics. **(4) Cross-asset robustness:** model rankings and a shared asset difficulty gradient are consistent across all 7 assets spanning cryptocurrency and US equity markets, indicating that the observed capability hierarchy is environment-agnostic. Taken together, these findings validate the strategy code-generation paradigm as a stable, reproducible, and discriminative framework for benchmarking LLM capabilities in quantitative finance. They also demonstrate that real-world queries alone can reveal meaningful capability differences among frontier models, thereby providing a solid empirical foundation and directly motivating the more fine-grained, difficulty-stratified diagnostic evaluation in Stage 2.

4.3 Results on LLM-augmented Queries

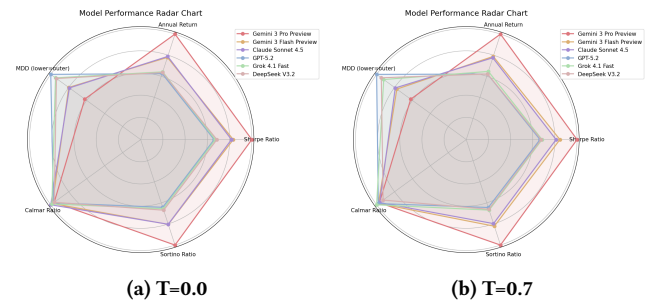


Figure 2: Multi-metric radar profiles on LLM-augmented queries (five metrics, $\tau=0$ vs. $\tau=0.7$). Near-identical polygons confirm temperature-invariant model behavior.

4.3.1 Overall model comparison. The Overall block of Table 2 reveals four key observations that extend and sharpen the Stage 1 findings. **(1) The three-tier hierarchy is preserved and amplified.** The same model ranking observed in Stage 1 recurs: *gemini-3-pro-preview* leads (SR = 0.628 at $T=0$), a middle tier follows (*gemini-3-flash-preview*, *claude-sonnet-4.5*), and the remaining models trail. Notably, the 7.8% ARR gap between the best and worst models exceeds the 5.5% gap in Stage 1, confirming that the controlled difficulty design of Stage 2 effectively amplifies latent capability differences among models. **(2) The return-risk inversion persists.** Consistent with Stage 1, return and risk rankings remain inversely correlated: the top return-generating model (*gemini-3-pro-preview*) incurs the highest drawdown and volatility, whereas the most conservative model (*gpt-5.2*, MDD = 0.119) trails on return metrics. *claude-sonnet-4.5* achieves the best Calmar Ratio (CR = 1.650 at $T=0$), exhibiting the most favorable return-to-drawdown trade-off, further reinforcing the necessity of multi-metric evaluation. **(3) Evaluation is temperature-invariant.** The $T=0$ and $T=0.7$ columns yield near-identical values across all models and metrics (max SR difference < 0.008). This invariance is a distinctive advantage of the code-generation paradigm: because downstream execution is deterministic, decoding temperature affects only surface-level code variation without altering the resulting trading logic, a property absent in direct-trading benchmarks where temperature directly perturbs each decision. **(4) Radar charts reveal distinct and stable risk profiles.** As shown in Figure 2, *gemini-3-pro-preview* spans the largest polygon, dominating on the return and Sortino axes while receding on MDD, visually encoding its aggressive strategy

Table 2: Results on LLM-augmented queries (mean $\pm$ std, 5 runs) at $T=0$ and $T=0.7$, stratified by overall (green), level (blue), and asset (gray). Best per block in bold; $\uparrow/\downarrow$: higher/lower is better.

Model	SR↑		ARR↑		MDD↓		CR↑		SOR↑		VOL↓	
	T=0	T=0.7	T=0	T=0.7	T=0	T=0.7	T=0	T=0.7	T=0	T=0.7	T=0	T=0.7
Overall												
claude-sonnet-4.5	0.513±0.270	0.508±0.266	0.164±0.114	0.162±0.113	0.150±0.111	0.143±0.110	1.650±0.864	1.634±0.620	0.806±0.478	0.795±0.471	0.205±0.153	0.202±0.152
deepseek-v3.2	0.430±0.280	0.424±0.289	0.132±0.110	0.130±0.112	0.127±0.108	0.123±0.109	1.586±0.800	1.570±0.761	0.672±0.477	0.660±0.489	0.173±0.149	0.168±0.150
gpt-5.2	0.415±0.307	0.417±0.308	0.130±0.122	0.130±0.121	0.119±0.116	0.117±0.114	1.599±0.794	1.660±0.821	0.645±0.525	0.643±0.522	0.163±0.160	0.160±0.157
gemin-3-flash-preview	0.523±0.289	0.530±0.264	0.162±0.122	0.165±0.113	0.148±0.117	0.151±0.111	1.618±0.904	1.658±1.384	0.807±0.506	0.808±0.468	0.204±0.162	0.206±0.153
gemin-3-pro-preview	0.628±0.255	0.627±0.242	0.208±0.112	0.209±0.107	0.191±0.109	0.188±0.104	1.586±0.665	1.639±0.651	1.004±0.465	0.999±0.439	0.262±0.150	0.259±0.143
grok-4.1-fast	0.421±0.269	0.429±0.267	0.134±0.108	0.135±0.107	0.125±0.102	0.127±0.102	1.629±0.741	1.692±0.819	0.658±0.461	0.668±0.457	0.171±0.140	0.173±0.140
Level 1 (Logic Translation)												
claude-sonnet-4.5	0.549±0.306	0.551±0.305	0.176±0.131	0.177±0.131	0.157±0.119	0.158±0.119	1.676±0.746	1.690±0.719	0.852±0.544	0.856±0.543	0.219±0.167	0.219±0.166
deepseek-v3.2	0.561±0.296	0.559±0.296	0.180±0.128	0.181±0.128	0.163±0.117	0.162±0.117	1.664±0.706	1.675±0.711	0.873±0.529	0.870±0.530	0.226±0.164	0.224±0.163
gpt-5.2	0.544±0.313	0.545±0.315	0.175±0.132	0.175±0.133	0.156±0.121	0.156±0.121	1.637±0.769	1.642±0.773	0.846±0.552	0.849±0.554	0.217±0.169	0.217±0.169
gemin-3-flash-preview	0.543±0.316	0.543±0.313	0.176±0.133	0.175±0.132	0.157±0.122	0.156±0.121	1.644±0.774	1.672±0.763	0.847±0.556	0.845±0.552	0.218±0.170	0.217±0.169
gemin-3-pro-preview	0.545±0.314	0.543±0.314	0.176±0.132	0.175±0.133	0.157±0.121	0.156±0.121	1.647±0.771	1.651±0.772	0.850±0.554	0.846±0.554	0.218±0.169	0.217±0.169
grok-4.1-fast	0.532±0.302	0.532±0.304	0.172±0.130	0.172±0.130	0.155±0.119	0.156±0.119	1.675±0.714	1.674±0.734	0.827±0.537	0.829±0.539	0.215±0.166	0.216±0.166
Level 2 (Parameter Inference)												
claude-sonnet-4.5	0.482±0.249	0.482±0.249	0.151±0.104	0.149±0.102	0.136±0.109	0.136±0.106	1.819±1.217	1.659±0.673	0.746±0.442	0.739±0.433	0.185±0.148	0.185±0.145
deepseek-v3.2	0.401±0.251	0.406±0.273	0.117±0.095	0.119±0.100	0.110±0.103	0.112±0.108	1.564±0.805	1.568±0.832	0.612±0.417	0.625±0.454	0.149±0.139	0.152±0.146
gpt-5.2	0.366±0.262	0.384±0.266	0.104±0.097	0.109±0.095	0.095±0.096	0.095±0.093	1.553±0.894	1.798±0.963	0.546±0.431	0.564±0.432	0.129±0.130	0.130±0.127
gemin-3-flash-preview	0.493±0.271	0.515±0.243	0.144±0.112	0.151±0.100	0.130±0.111	0.138±0.105	1.698±1.239	1.792±2.233	0.736±0.462	0.778±0.420	0.177±0.152	0.188±0.142
gemin-3-pro-preview	0.604±0.224	0.632±0.197	0.196±0.096	0.209±0.089	0.171±0.095	0.184±0.091	1.684±0.761	1.740±0.753	0.942±0.394	0.931±0.357	0.234±0.129	0.252±0.123
grok-4.1-fast	0.401±0.240	0.422±0.239	0.127±0.090	0.133±0.089	0.117±0.088	0.122±0.089	1.639±0.803	1.820±1.084	0.622±0.402	0.653±0.395	0.158±0.119	0.165±0.120
Level 3 (Goal-Oriented Generation)												
claude-sonnet-4.5	0.507±0.247	0.492±0.234	0.165±0.104	0.160±0.102	0.156±0.102	0.148±0.104	1.452±0.353	1.553±0.414	0.820±0.434	0.792±0.420	0.212±0.140	0.202±0.142
deepseek-v3.2	0.329±0.240	0.304±0.235	0.099±0.086	0.090±0.084	0.107±0.094	0.095±0.087	1.530±0.876	1.458±0.723	0.531±0.407	0.483±0.391	0.142±0.126	0.126±0.118
gpt-5.2	0.336±0.303	0.322±0.297	0.112±0.120	0.106±0.118	0.105±0.120	0.099±0.114	1.608±0.702	1.542±0.682	0.542±0.524	0.516±0.510	0.143±0.163	0.135±0.156
gemin-3-flash-preview	0.532±0.276	0.531±0.226	0.167±0.119	0.169±0.102	0.159±0.116	0.158±0.105	1.511±0.541	1.509±0.347	0.838±0.487	0.837±0.416	0.216±0.159	0.214±0.144
gemin-3-pro-preview	0.734±0.167	0.705±0.157	0.252±0.087	0.244±0.081	0.245±0.086	0.225±0.082	1.429±0.341	1.526±0.293	1.221±0.335	1.159±0.309	0.334±0.118	0.309±0.114
grok-4.1-fast	0.331±0.219	0.332±0.211	0.102±0.085	0.100±0.082	0.105±0.088	0.102±0.086	1.573±0.701	1.578±0.510	0.526±0.375	0.521±0.362	0.140±0.119	0.137±0.117
Bitcoin (Cryptocurrency)												
claude-sonnet-4.5	0.353±0.186	0.354±0.186	0.134±0.085	0.133±0.085	0.203±0.128	0.199±0.128	0.817±0.655	0.833±0.676	0.601±0.330	0.601±0.327	0.233±0.149	0.230±0.149
deepseek-v3.2	0.307±0.202	0.297±0.205	0.114±0.087	0.110±0.088	0.173±0.130	0.169±0.131	0.821±0.687	0.854±0.758	0.526±0.348	0.513±0.348	0.198±0.150	0.194±0.152
gpt-5.2	0.301±0.212	0.302±0.211	0.111±0.094	0.111±0.093	0.158±0.139	0.159±0.137	0.958±0.781	0.954±0.789	0.497±0.373	0.499±0.370	0.184±0.161	0.185±0.159
gemin-3-flash-preview	0.373±0.205	0.371±0.176	0.137±0.094	0.138±0.085	0.203±0.139	0.205±0.130	0.876±0.723	0.855±0.683	0.621±0.357	0.624±0.315	0.235±0.160	0.238±0.150
gemin-3-pro-preview	0.420±0.162	0.425±0.143	0.164±0.078	0.167±0.071	0.254±0.125	0.251±0.118	0.797±0.643	0.837±0.623	0.739±0.313	0.736±0.258	0.294±0.142	0.291±0.134
grok-4.1-fast	0.299±0.194	0.313±0.187	0.111±0.086	0.114±0.084	0.174±0.124	0.177±0.124	0.797±0.715	0.869±0.710	0.508±0.334	0.524±0.325	0.199±0.142	0.203±0.142
Ethereum (Cryptocurrency)												
claude-sonnet-4.5	0.297±0.225	0.290±0.226	0.151±0.150	0.147±0.150	0.245±0.184	0.243±0.183	0.592±0.376	0.563±0.346	0.465±0.348	0.455±0.349	0.285±0.220	0.282±0.220
deepseek-v3.2	0.252±0.218	0.244±0.217	0.123±0.138	0.118±0.137	0.216±0.182	0.211±0.184	0.545±0.370	0.560±0.440	0.396±0.336	0.382±0.335	0.249±0.216	0.243±0.218
gpt-5.2	0.233±0.241	0.229±0.236	0.121±0.153	0.118±0.152	0.197±0.193	0.194±0.191	0.564±0.405	0.562±0.365	0.369±0.370	0.361±0.363	0.228±0.230	0.224±0.227
gemin-3-flash-preview	0.276±0.260	0.292±0.230	0.141±0.162	0.145±0.152	0.246±0.195	0.249±0.184	0.507±0.416	0.544±0.354	0.444±0.385	0.460±0.352	0.284±0.235	0.288±0.222
gemin-3-pro-preview	0.358±0.224	0.355±0.215	0.181±0.156	0.179±0.149	0.311±0.174	0.305±0.166	0.543±0.360	0.559±0.330	0.569±0.341	0.564±0.326	0.360±0.211	0.354±0.202
grok-4.1-fast	0.245±0.213	0.248±0.211	0.121±0.135	0.122±0.135	0.214±0.174	0.218±0.172	0.529±0.363	0.515±0.340	0.387±0.327	0.393±0.322	0.247±0.207	0.251±0.205
AAPL (US Equity)												
claude-sonnet-4.5	0.941±0.493	0.924±0.488	0.180±0.119	0.175±0.117	0.073±0.048	0.071±0.047	2.799±1.303	2.779±1.330	1.262±0.726	1.240±0.721	0.116±0.074	0.113±0.072
deepseek-v3.2	0.803±0.515	0.784±0.526	0.145±0.116	0.142±0.119	0.059±0.047	0.058±0.048	2.774±1.499	2.722±1.400	1.074±0.732	1.047±0.750	0.095±0.072	0.093±0.074
gpt-5.2	0.745±0.549	0.751±0.548	0.139±0.125	0.139±0.125	0.058±0.052	0.058±0.052	2.621±1.493	2.745±1.577	0.977±0.780	0.982±0.778	0.092±0.079	0.091±0.079
gemin-3-flash-preview	0.946±0.505	0.952±0.472	0.173±0.127	0.173±0.117	0.087±0.064	0.084±0.063	2.636±1.621	2.648±1.621	1.055±0.897	1.051±0.890	0.131±0.134	0.127±0.131
gemin-3-pro-preview	1.067±0.480	1.075±0.446	0.214±0.123	0.215±0.115	0.093±0.049	0.091±0.046	2.539±1.472	2.661±1.312	1.456±0.729	1.462±0.679	0.146±0.073	0.144±0.070
grok-4.1-fast	0.744±0.494	0.751±0.485	0.139±0.114	0.139±0.112	0.059±0.046	0.059±0.046	2.613±1.453	2.615±1.386	0.986±0.711	0.993±0.697	0.094±0.071	0.094±0.070
GOOGL (US Equity)												
claude-sonnet-4.5	0.792±0.445	0.789±0.434	0.201±0.146	0.195±0.140	0.103±0.089	0.103±0.089	2.575±1.566	2.578±1.596	1.316±0.823	1.308±0.799	0.168±0.132	0.168±0.132
deepseek-v3.2	0.628±0.455	0.627±0.470	0.146±0.138	0.146±0.142	0.087±0.064	0.084±0.063	2.261±1.689	2.350±1.720	1.044±0.815	1.036±0.837	0.139±0.131	0.136±0.132
gpt-5.2	0.635±0.502	0.640±0.502	0.160±0.154	0.161±0.154	0.080±0.088	0.078±0.086	2.636±1.621	2.648±1.621	1.051±0.890	1.051±0.890	0.131±0.134	0.127±0.131
gemin-3-flash-preview	0.802±0.492	0.808±0.442	0.195±0.157	0.199±0.146	0.104±0.097	0.105±0.089	2.435±1.751	2.487±1.605	1.325±0.902	1.336±0.822	0.170±0.143	0.170±0.132
gemin-3-pro-preview	0.960±0.446	0.968±0.423	0.252±0.153	0.256±0.143	0.131±0.097	0.130±0.090	2.633±1.765	2.700±1.552	1.627±0.834	1.644±0.789	0.212±0.140	0.211±0.131
grok-4.1-fast	0.639±0.446	0.644±0.440	0.157±0.138	0.157±0.137	0.082±0.082	0.084±0.084	2.646±1.915	2.830±2.808	1.065±0.798	1.079±0.785	0.133±0.123	0.137±0.125
MSFT (US Equity)												
claude-sonnet-4.5	0.438±0.312	0.433±0.308	0.084±0.062	0.083±0.061	0.078±0.047	0.077±0.046	1.145±1.180	1.097±0.971	0.657±0.450	0.644±0.443	0.113±0.067	0.111±0.065
deepseek-v3.2	0.362±0.314	0.360±0.321	0.065±0.064	0.064±0.062	0.062±0.046	0.061±0.045	1.062±0.975	1.035±0.975	0.529±0.455	0.525±0.469	0.088±0.068	0.088±0.068
gpt-5.2	0.376±0.336	0.376±0.333	0.070±0.064	0.070±0.064	0.060±0.050	0.062±0.050	1.130±0.937	1.151±1.010	0.546±0.469	0.543±0.480	0.092±0.071	0.091±0.073
gemin-3-flash-preview	0.474±0.339	0.475±0.306	0.089±0.067	0.090±0.062	0.079±0.050	0.080±0.046	1.147±0.918	1.037±2.709	0.692±0.498	0.696±0.439	0.115±0.070	0.117±0.064
gemin-3-pro-preview	0.576±0.31											

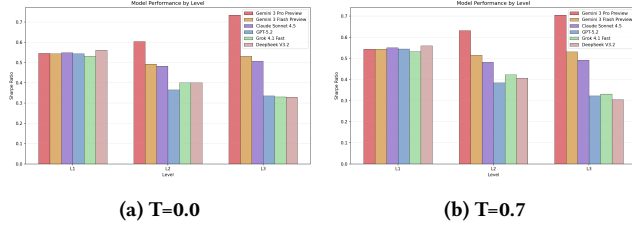


Figure 3: Sharpe Ratio by difficulty level on LLM-augmented queries ($\tau=0$ vs. $\tau=0.7$). Inter-model spread widens monotonically from Level 1 to Level 3.

thresholds, lookback windows, and indicator parameters, indicating that grounding underspecified strategy skeletons in reasonable financial parameters effectively separates models with strong domain knowledge from those that lack it; stochastic sampling ($\tau=0.7$) marginally benefits the top model at this level, suggesting that sampling diversity can occasionally discover better parameter configurations. **(3) Goal-oriented generation (Level 3) produces the most discriminative separation:** the spread reaches 14 $\times$ the Level 1 range, with a striking crossover where models that excel at constrained translation (e.g., *deepseek-v3.2*) decline monotonically from L1 to L3, whereas *gemini-3-pro-preview* actually improves, indicating superior open-ended strategy design capabilities. The primary cognitive leap occurs between Level 1 and Level 2 (a 16% SR decline in model-averaged performance), while the L2-to-L3 transition introduces additional inter-model variance rather than a sharp further mean decline. This crossover demonstrates that the three levels probe fundamentally different cognitive capabilities, and no single model dominates across all levels.

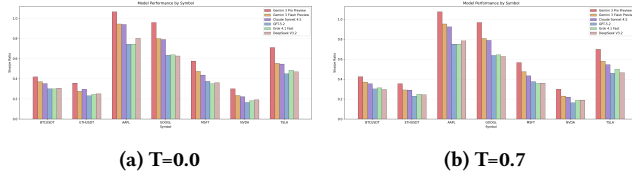


Figure 4: Sharpe Ratio by asset on LLM-augmented queries ($\tau=0$ vs. $\tau=0.7$). A shared difficulty gradient emerges, with AAPL/GOOGL easiest and crypto/NVDA hardest.

4.3.3 Per-asset analysis. The asset blocks of Table 2 and Figure 4 yield four observations that mirror and extend the Stage 1 per-asset findings. **(1) A systematic difficulty gradient persists across assets.** Trend-rich large-caps (AAPL, GOOGL) remain the easiest targets for all models, MSFT and NVDA constitute the hardest equity environments primarily due to narrower trading ranges and rapid regime shifts respectively, and cryptocurrencies (BTCUSDT, ETHUSDT) rank as the most challenging overall, reflecting higher volatility and fundamentally different 24/7 market dynamics. TSLA again occupies a unique position: it yields the highest absolute returns yet the widest cross-model variance, amplifying the advantage of aggressive signal logic. **(2) The Stage 2 difficulty design sharpens inter-model dispersion within each asset.** Compared to Stage 1, the controlled query design produces wider SR spreads on every asset, confirming that the structured queries effectively magnify latent capability differences that real-world queries alone partially obscure. **(3) Distinct model specialization**

patterns emerge. The top return-generating model excels particularly on high-volatility assets (TSLA, NVDA, cryptocurrencies), suggesting robust handling of challenging market conditions; conversely, *claude-sonnet-4.5* exhibits the most uniform performance across assets, indicating balanced, asset-agnostic strategy generation. **(4) Rankings and difficulty gradients are temperature-invariant.** The bar patterns visualized in Figure 4 remain virtually identical between the $\tau=0$ and $\tau=0.7$ panels, confirming that both model specialization profiles and the asset-difficulty hierarchy are intrinsic properties fundamentally unaffected by decoding temperature. Detailed per-asset tables and additional visual analyses are provided in Appx. G.

4.3.4 Aligned return curve analysis. Figure 5 overlays the cumulative return trajectories of all six models across the full query spectrum, with shaded bands denoting the 25th–75th percentile range over 5 independent runs. Four observations emerge. **(1) Inter-model separation is persistent and substantial.** The vertical gap between the highest and lowest trajectories far exceeds any individual model’s confidence band throughout the entire query range, providing direct visual evidence that performance differences reflect genuine capability gaps rather than generation noise. **(2) Run-to-run stability is consistently high.** The narrow shaded bands confirm that independently generated strategies produce tightly clustered outcomes even at $\tau=0.7$; notably, *claude-sonnet-4.5* exhibits the narrowest bands (most consistent generation), while *grok-4.1-fast* displays the widest, corroborating its high-variance profile observed in the boxplot analysis. **(3) Temperature invariance is visually confirmed.** The $\tau=0$ and $\tau=0.7$ panels yield virtually identical curve shapes and separation patterns, reinforcing that the code-generation paradigm confines LLM stochasticity to a single generation step while ensuring deterministic downstream execution. **(4) Difficulty modulates inter-model divergence.** The separation widens progressively for harder queries and narrows for easier ones, consistent with the per-level finding that Level 3 tasks maximally amplify capability differences across models.

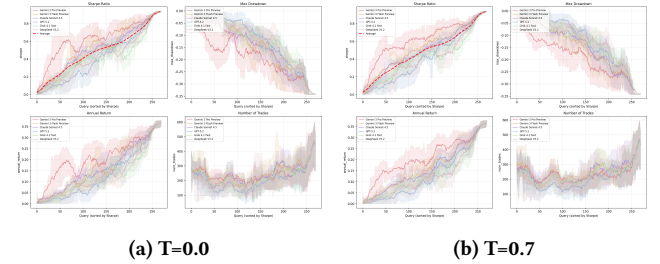


Figure 5: Aligned cumulative return curves on LLM-augmented queries ($\tau=0$ vs. $\tau=0.7$). Shaded bands denote the 25th–75th percentile range over 5 runs.

4.3.5 Per-model stability profiles. Appx. G provides detailed per-model analyses of multi-run stability, difficulty-level performance curves, and best-case strategy returns. Three archetypal behavioral profiles emerge from these results. **(1) Aggressive-creative profile (*gemini-3-pro-preview*):** uniquely, this model’s SR increases from Level 1 to Level 3, indicating that open-ended creative freedom amplifies its strengths; however, this comes at the cost of the

widest confidence bands among top-tier models and the highest drawdown/volatility. **(2) Balanced-stable profile** (*claude-sonnet-4.5*, *gemini-3-flash-preview*): these models exhibit the narrowest run-to-run confidence bands and the mildest Level 1-to-Level 3 degradation, with *claude-sonnet-4.5* achieving the best Calmar Ratio (CR = 1.650) and *gemini-3-flash-preview* serving as a cost-effective alternative with a nearly flat difficulty profile. **(3) Conservative-rigid profile** (*gpt-5.2*, *deepseek-v3.2*, *grok-4.1-fast*): these models favor low-risk signal logic (lowest MDD and VOL) but suffer the steepest performance degradation from structured to open-ended tasks; *deepseek-v3.2* exemplifies this pattern most starkly, leading at Level 1 yet dropping to the bottom tier at Level 3, while *grok-4.1-fast* additionally exhibits the highest run-to-run variance, making it the least predictable model in the benchmark. These archetypal profiles suggest that the return-risk trade-off observed in aggregate metrics stems from fundamentally different strategy-generation strategies encoded by each LLM, and that no single model simultaneously optimizes for return, stability, and robustness across difficulty levels.

4.3.6 Findings and conclusions. Synthesizing the above analyses, the Stage 2 evaluation yields five principal findings. **(1) Temperature invariance:** the $\tau=0$ and $\tau=0.7$ results are near-identical across models, metrics, and difficulty levels, with a maximum SR difference below 0.008. **(2) Systematic difficulty progression:** the inter-model SR spread widens from Level 1 (0.029) to Level 2 (8 $\times$) and Level 3 (14 $\times$), confirming that the 3×3 taxonomy separates models along a controlled cognitive-demand axis. **(3) Cross-level ranking reversals:** Level 1 code translation and Level 3 open-ended strategy design rely on distinct capabilities, as shown by the strong crossover in model rankings that would be obscured by aggregate metrics and single-score summaries. **(4) Reproducible risk profiles:** the aggressive-creative, balanced-stable, and conservative-rigid patterns persist across stages, assets, and temperatures, indicating stable model-specific strategy behavior under both structured and open-ended query settings. **(5) Cross-asset robustness:** model rankings and a shared asset-difficulty gradient are preserved across all 7 assets spanning cryptocurrency and US equity markets, with Stage 2 further sharpening inter-model dispersion within each asset. Together, these findings validate the 3×3 taxonomy as an effective diagnostic tool while complementing the ecological validity of Stage 1, and reinforce the code-generation paradigm as a stable, reproducible, and discriminative benchmark for LLM capability evaluation inference across both real-world and structured queries.

5 Discussion

The results suggest that strategy code generation changes both the evaluation interface and the capability under study. Here, end to end denotes the full pipeline from natural language to complete rule based strategy generation and deterministic backtesting, rather than agent style systems that directly emit BUY, HOLD, or SELL actions. The two settings therefore probe different abilities. Agent based trading emphasizes step wise action selection, whereas **ALPHAForgeBENCH** focuses on whether a model can synthesize factors, rules, and decision logic into a coherent strategy. The near invariance between $\tau=0$ and $\tau=0.7$ and the low run to run variance across both stages indicate that, once execution randomness is removed, performance differences more cleanly reflect strategy

design ability. This also explains why agent style financial trading is a weak benchmark here, since Appendix C shows that such systems can remain unstable even under identical configurations.

The results further show that financial strategy generation is not a single capability. The widening spread across difficulty levels and the ranking reversals between Level 1 and Level 3 indicate that logic translation, logic completion, and open ended synthesis rely on distinct strengths. The recurring risk profiles across stages and assets likewise suggest that frontier LLMs encode stable preferences in how they trade off return, drawdown, and volatility. **ALPHAForgeBENCH** is therefore useful not only as a leaderboard but also as a diagnostic benchmark that separates conservative from aggressive models, structured translators from creative synthesizers, and robust models from those that degrade as task ambiguity increases. This also motivates the current focus on single asset evaluation, which isolates signal generation and strategy logic construction without the added variability of asset allocation, cross asset dependencies, and risk control.

The benchmark should be interpreted within a controlled scope. The current evaluation targets single asset long only strategy design under a standardized backtesting environment with fixed transaction cost of 10^{-3} and without explicit slippage or liquidity modeling. These choices improve comparability and reproducibility, but they also mean that the reported results are better read as controlled measures of strategy design quality than as direct estimates of deployable trading performance. The same logic applies to Stage 2, whose queries are generated through controlled augmentation of Stage 1 to provide diagnostic structure rather than broad ecological coverage. Because Stage 2 remains anchored to real world sources and the predefined 3×3 taxonomy, and model rankings remain broadly aligned between Stage 1 and Stage 2, the structured track appears to preserve genuine capability differences without strong model specific bias. Under this interpretation, **ALPHAForgeBENCH** provides a stable and interpretable foundation for evaluating financial strategy design with LLMs.

6 Conclusion

We presented **ALPHAForgeBENCH**, a benchmark that reframes LLM evaluation in quantitative finance from black box action emission to white box strategy code generation. Across 903 queries, six frontier LLMs, seven assets, and 35,190 total implementations, the results show that this code generation paradigm is temperature invariant, highly reproducible, and more discriminative than direct trading baselines. The benchmark further reveals widening capability gaps across difficulty levels, ranking reversals that separate distinct cognitive skills, and stable model specific risk profiles, establishing **ALPHAForgeBENCH** as a rigorous framework for evaluating financial strategy design with LLMs.

7 Acknowledgments

Shuo Sun is supported by Guangdong Provincial Key Lab of Integrated Communication, Sensing and Computation for Ubiquitous Internet of Things (No.2023B1212010007). Bo An is supported by the National Research Foundation Singapore and DSO National Laboratories under the AI Singapore Programme (AISG Award No: AISG2-GC-2023-009-1B).

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